

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: CRYPTOGRAPHY AND NETWORK
SECURITY COURSE CODE: CSA5111**

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Kala Gowri Shankar (192325109) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

**Annexure A
Scenario-based**

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

Hellman key exchange works

Diffie–Hellman (DH) is a method that allows two users to create a shared secret key over a public network without sending the secret key directly.

Steps:

1. Two public values are chosen:
 - Prime number **P**
 - Generator **G**
2. **User A** chooses a private key **a** and computes:

User A sends **A** to User B.

3. **User B** chooses a private key **b** and computes:

User B sends **B** to User A.

4. User A calculates:

5. User B calculates:

6. Both users obtain the **same shared secret key (K)**, which is then used for secure communication.

Example

Given:

- Prime number **P = 23**
- Generator **G = 5**
- User A private key **a = 6**
- User B private key **b = 15**

Calculate public keys:

User A:

User B:

Shared key:

User A:

User B:

Shared Secret Key = 2

B. Identify a possible security threat and suggest a solution

Security Threat

Man-in-the-Middle (MITM) Attack

- An attacker intercepts the exchanged public keys.
- The attacker establishes separate keys with both users.
- Both users believe they are communicating securely, but the attacker can read and modify all messages.

Solution

- Use **Digital Certificates** and **RSA-based Digital Signatures** to authenticate public keys.
- Implement **TLS/SSL** with certificate verification.
- Verify the identity of both users before accepting public keys.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

A. Describe how RSA is used for encryption and decryption

RSA uses two keys:

- **Public Key** → Encrypts data
- **Private Key** → Decrypts data

Encryption Process

1. Every user has a public key and a private key.
2. The sender encrypts the email using the receiver's **public key**.
3. The encrypted email is sent over the network.
4. Only the receiver can decrypt it using their **private key**.

Example

Suppose:

- Public Key = (e, n)
- Private Key = (d, n)

Encryption:

where:

- M = Message
- C = Ciphertext

Decryption:

Only the receiver's private key can recover the original message.

B. Explain how key distribution and management maintain security

Key Distribution

- Public keys are shared openly.
- Private keys are kept secret.
- Public keys are distributed using **Digital Certificates** issued by a trusted **Certificate Authority (CA)**.

Key Management

Good key management includes:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Secure generation of keys. Safe storage of private keys. Periodic key rotation. Revoking compromised keys. Using strong key lengths (2048-bit or higher).	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
	25%	Applies	Applies	Limited or	Unable to

Application

of Concepts		appropriate concepts effectively to address the scenario	concepts with minor errors	partially correct application	apply relevant concepts
Decision Making	20%	Makes well reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justificatio n	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganiz ed and difficult to understand

RUBRICS FOR CALCULATION: